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Assignment No. 4 (Part II)

Title:

Implement DEAP (Distributed Evolutionary Algorithms) using Python.

Theory:

Distributed Evolutionary Algorithms (DEAP):

- Distributed Evolutionary Algorithms are a class of evolutionary algorithms designed to solve complex optimization problems by distributing computation across multiple processors or computational nodes.
- DEAP offers scalability, efficiency, and parallelism, making it suitable for solving large-scale optimization problems.

DEAP Components:

- Population: A collection of candidate solutions (individuals) representing potential solutions to the optimization problem.
- Selection: The process of selecting individuals from the population for reproduction based on their fitness.
- Crossover: The process of combining genetic material from selected individuals to create new offspring.
- Mutation: The process of introducing random changes to the genetic material of individuals to maintain diversity.
- Evaluation: The process of evaluating the fitness of individuals in the population based on their performance on the optimization problem.
- Termination Criteria: The condition under which the optimization process terminates, such as reaching a maximum number of generations or achieving a desired fitness level.

Features of DEAP:

- Parallelism: DEAP utilizes parallel processing to evaluate fitness functions and explore the search space efficiently.
- Flexibility: DEAP provides a flexible framework for implementing various evolutionary algorithms, including Genetic Algorithms, Evolutionary Strategies, and Genetic Programming.
- Customization: DEAP allows customization of operators, selection methods, and termination criteria to tailor the algorithm to specific problem domains.

Conclusion: Thus, we have successfully implemented a program for DEAP.